

Step 0: Initial Document Check

- **0.1:** The document is an RCC structural drawing of "FOUNDATIONS" only.
- **0.2:** Site location is not mentioned.
- **0.3:** All compliance checks will be based only on IS 456:2000 and SP 34.

Step 1: Locate the "NOTES" Section

- **Status:** Compliant

Step 2 & 3: Extract and Verify Design Parameters from "NOTES"

Criteria	Extracted Value	Compliance Check	Status
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1. Grade of Concrete	M20 for all R.C.C works	M20 is a common grade. Suitability for environmental exposure cannot be verified without site location.	Cannot Verify
2. Reinforcement Bars	HYSD TMT bars of Grade Fe 500 conforming to IS 1786-2008. Shear reinforcement and stirrups are mentioned in the "SCHEDULE OF COLUMNS" as "TIES Y8@8" and "TIES Y8@4".	Fe 500 is standard. The mention of stirrups/ties is good.	Compliant
3. Lap Length	50 times dia of the bar, to be staggered such that not more than 50% of the bars are lapped at a section.	Complies with the general rule of 50d. Staggering is good practice.	Compliant
4. Clear Cover	Footing/Wall: 50 mm, Columns: 40 mm, Beam: 25 mm, Slab: 20 mm.	Footing cover (50mm) is minimum. PCC is mentioned in Section X, which is good for a 50mm cover. Column cover (40mm) is within the 40-70mm range. Beam and Slab covers are typical.	Compliant
5. Development Length (Ld)	50 times the dia of the bar.	A common value (50d).	Compliant
6. Safe Bearing Capacity (SBC) of soil	18 T/m ²	SBC is mentioned. PCC is specified in Section X.	Compliant
7. Seismic Zone and Wind Load	Not mentioned.	This information is mandatory for structural design.	Missing Information
8. Building Limitations	Structure is designed for G+1 storey.	Building limitation (G+1 storey) is mentioned.	Compliant
9. Structure's Purpose	Not explicitly mentioned, but implied as a G+1 storey building.	The specific purpose (e.g., residential, commercial) is not stated.	Missing Information
10. Floor Heights	Not mentioned.	Floor heights are not specified.	Missing Information
11. Schedule of Footings	A "SCHEDULE OF FOOTINGS" table is present and appears consistent with the "FOUNDATION PLAN".	The table is present and seems consistent.	Compliant

12. Footing Type	Isolated footings (F1 to F5).	Isolated footings are used for a G+1 storey building, which is appropriate for low-rise structures.	Compliant
13. Reinforcement in High-Rise Buildings	Not applicable as the structure is G+1 storey (low-rise).	Not Applicable	Not Applicable
14. Raft Foundation Reinforcement	Not applicable as isolated footings are used.	Not Applicable	Not Applicable
15. Lift Design	Not mentioned, and no lift pit is shown.	Not Applicable	Not Applicable
16. Soil Improvement	Not mentioned.	No details about soil improvement are provided.	Missing Information
17. Column Ties	"COLUMN TIES" are indicated in Section X and "TIES" are specified in the "SCHEDULE OF COLUMNS". They appear continuous in the typical column details.	Column ties are shown and specified.	Compliant
18. Plan of Ties	A separate plan for ties is not explicitly present, but tie details are given in the "SCHEDULE OF COLUMNS".	While not a separate plan, the details are provided.	Compliant
19. Outer Ties Check	Column C1: Main bars 8Y12, Ties Y8@8. Column C2: Main bars 4Y16 + 4Y12, Ties Y8@8. Column C3: Main bars 4Y16 + 4Y12, Ties Y8@4. Percentage of steel cannot be calculated without column dimensions.	The number and diameter of bars and ties are specified. Percentage of steel cannot be verified without column dimensions.	Cannot Verify
20. Cross-Section Area	Not explicitly specified whether gross or effective area is used.	This information is not explicitly stated in the notes.	Missing Information
21. Steel Curtailment	Not applicable for a G+1 storey building.	Not Applicable	Not Applicable
22. Maximum Steel Percentage in Columns	Not mentioned in the notes.	The maximum percentage of steel is not specified.	Missing Information

Step 5: Report Missing or Wrong Information

Site location is not mentioned.

Grade of Concrete suitability for environmental exposure cannot be verified without site location.

Seismic Zone and Wind Load are not mentioned.

Structure's Purpose is not explicitly stated.

Floor Heights are not mentioned.

Soil Improvement details are not mentioned.

Percentage of steel in columns cannot be verified without column dimensions.

Cross-section area usage (gross/effective) is not specified.

Maximum Steel Percentage in Columns is not mentioned.

Summary of Compliance

• **Total Criteria Evaluated:** 22

• **Compliant Items:** 8

- **Non-Compliant Items:** 0

- **Missing Information Items:** 7

- **Overall Verdict:** The drawing has several missing pieces of information that are crucial for a complete compliance check, particularly regarding site-specific data and some design assumptions. While the foundation plan itself is present, the lack of these details makes a full compliance assessment difficult.